CAST notes

A new mindset for incident analysis

Samuele Tonon - 20/03/2025

Booking.com

RCA, Post-Mortem, AAR, PIA, RFO...

- SRE in IT engineering is famous for their incident analysis methodologies.
- Incident analysis as an important step to learn - the hard way - valued lessons.
- What if there was already similar role outside of IT ?
- What if we could learn a thing or two from such discipline?

...Safety engineering enters the chat.

Nancy G. Leveson

- Professor of Aeronautics and Astronautics MIT.
- 2020 IEEE Medal for Environmental and Safety Technologies for her development of STAMP and other system safety and accident modeling analysis tools.
- 200+ papers on safety engineering.
- Main Author of CAST, STPA and STAMP.
- "Safeware: System Safety and Computers" book.
- "Engineering a Safer World" book.

;login:

The Evolution of SRE at Google

Using STAMP to improve resilience in Google production systems

December 18, 2024

RESEARCH

Authors: Tim Falzone, Ben Treynor Sloss

Article shepherded by: Rik Farrow

Billions of people around the world use Google's products every day, and they count on those products to work reliably. Behind the scenes, Google's services have increased dramatically in scale over the last 25 years — and failures have become rarer even as the scale has grown. Google's SRE team has pioneered methods to keep failures rare by engineering reliability into every part of the stack. SREs have scaled up methods that have gotten us very far—Service Level

What is CAST?

- Causal Analysis based on System Theory (CAST)
  - Goal is to maximizes learning from **accidents** and **incidents**.

- Technique used in **safety engineering**
  - Analysis of accidents happening in aerospace, military, oil and gas industries

- CAST try to approach accident analysis in a different way than the traditional approach
  - CAST follows the causality model of STAMP

Accidents vs Incidents

- Accidents
  - unexpected events that result in personal injury, property damage, illness, or **death**.
- Incidents
  - unexpected events that have the **potential** to result in injury, damage....

Traditional approach

Accidents are the result of a chain of events, where the events usually involve failures of system components and each failure or event is the direct cause of the one preceding it.

SOLUTION

Evaluate reliability of single components separately and later combine analysis results in to a system reliability value.

- Redundancy
- High component integrity and overdesign
- Fail-safe design
- Operational procedures, checklist, trainings


IF COMPONENTS DO NOT FAIL
THEN ACCIDENTS WON'T OCCUR.
RIGHT?
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A complex world

- The traditional approach no longer works in a complex system
    - software components behaving correctly but creating unexpected scenarios

- Chain of events unpredictable due to lack of linearity
    - If we assume A -> B and B does not occur than A didn't happen
    - Tobacco Lobby used this for a long time
    - *if "smoking causes lung cancer" how come there are smokers with no lung cancer?*

- We know there's a connection...
    - but it's not simple or direct

Case study: DLH 2904 to Warsaw (1993)

- Tower informs airplane of crosswind during landing phase .
- Pilots perform one wheel-first touchdown as per theory for crosswind landings (so the wind will push down the plane and the second wheel).
- Wind changed to tailwind and due to rain in the runway the only wheel touching, went aquaplaning for 9 seconds.
- Reverse thrust can only activate if one of the wheels is turning or there's weight on both of them.
- Due to aquaplaning the only wheel touching the ground did not spin for 9 seconds so reverse thrust could not be activated in time.
- The airplane was not able to stop before the end of the runway.
- All components worked as expected, no human error reported.

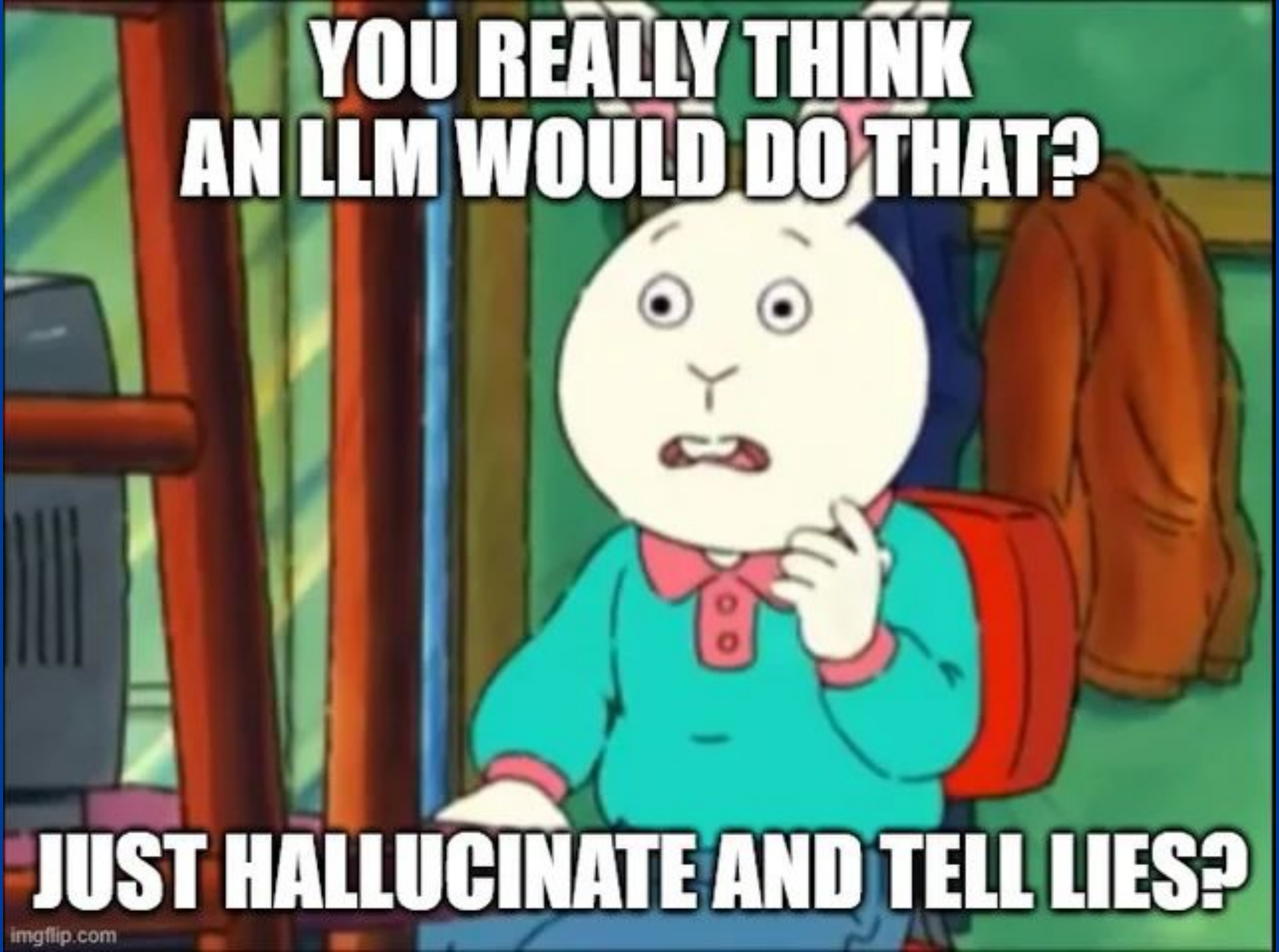
TikToker says she was trapped in 115-degree heat after her Tesla locked her inside during a software update

Katherine Tangalakis-Lippert Apr 13, 2024, 6:03 AM CEST

Share Save

A TikToker says she got stuck in her Tesla for 40 minutes during a software update. Brandon Bell/Getty Images

Insert random incident with AI


YOU REALLY THINK
AN LLM WOULD DO THAT?
JUST HALLUCINATE AND TELL LIES?
imgflip.com


IT'S NOT A LIE
IF YOU BELIEVE IT!
imgflip.com

STAMP as a new approach

- System-Theoretic Accident Model and Processes
  - Based on system theory not reliability theory
- No assumption of direct causality
  - E_1 did not necessarily cause E_2 or a chain of events such as E_2,E_3,E_4,E_5
- Treat accidents as a *holistic* system <u>*control*</u> problem, not a *failure* problem
  - Include software, humans, organisation, culture etc. as causal factors

Enforce a SAFETY through constraints on system behaviour

Safety through constraints on system behaviour

- Two planes/cars must not violate minimum separation.
- Toxic chemical/radiation must not be released from the plant.
- Weapons must not target friendly forces.
- Payments must be safe and secure.
- All input data in an IT system must be validated and sanitized

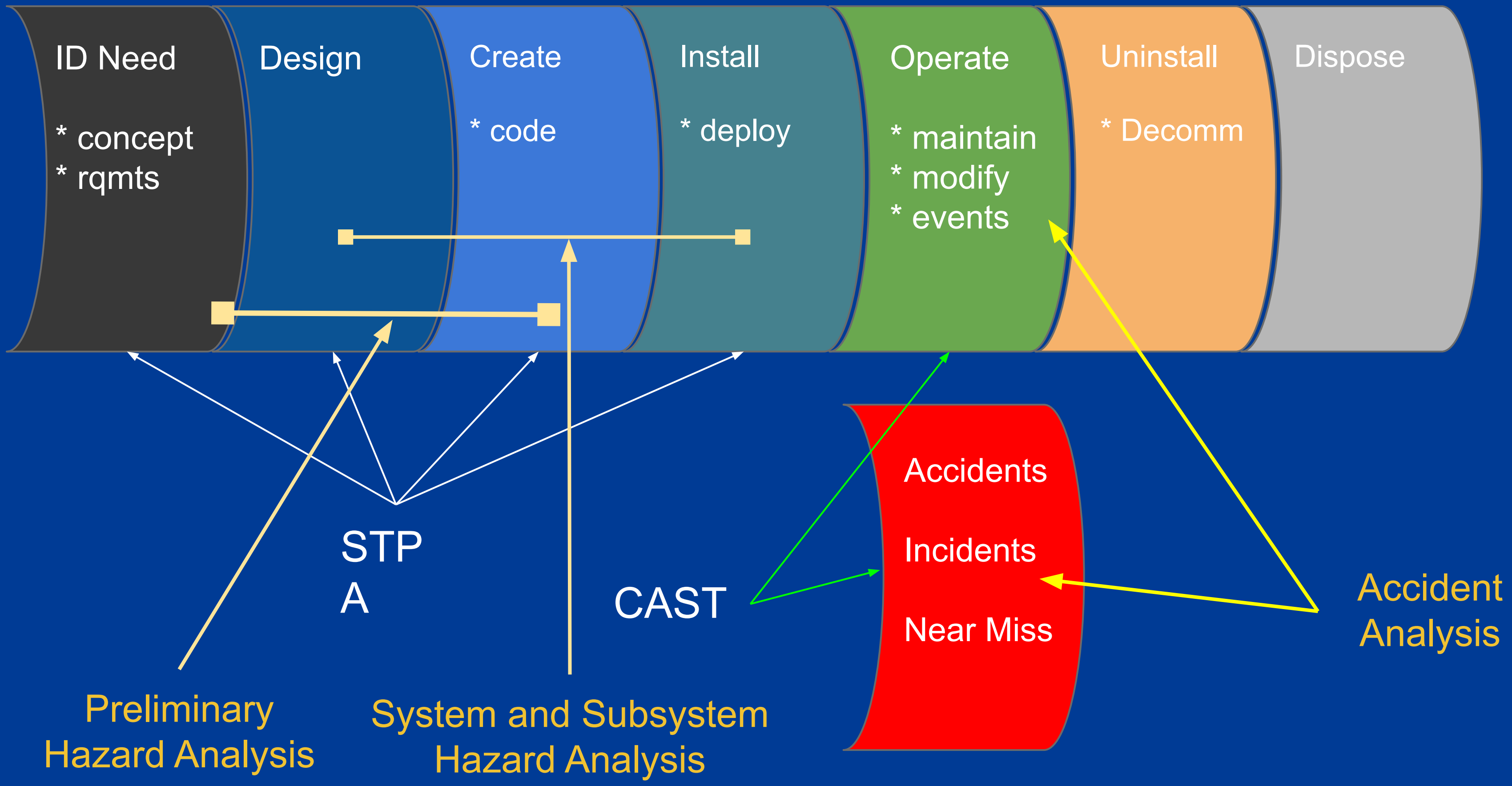
System Theoretic ecosystem

Analysis framework that can be applied during hazard analysis

- System Theoretic Process Analysis (STPA)
- System Theoretic Process Analysis - Security (STPA-Sec)

Analysis framework to be applied at accident analysis

- Causal Analysis based on System Theory (CAST)

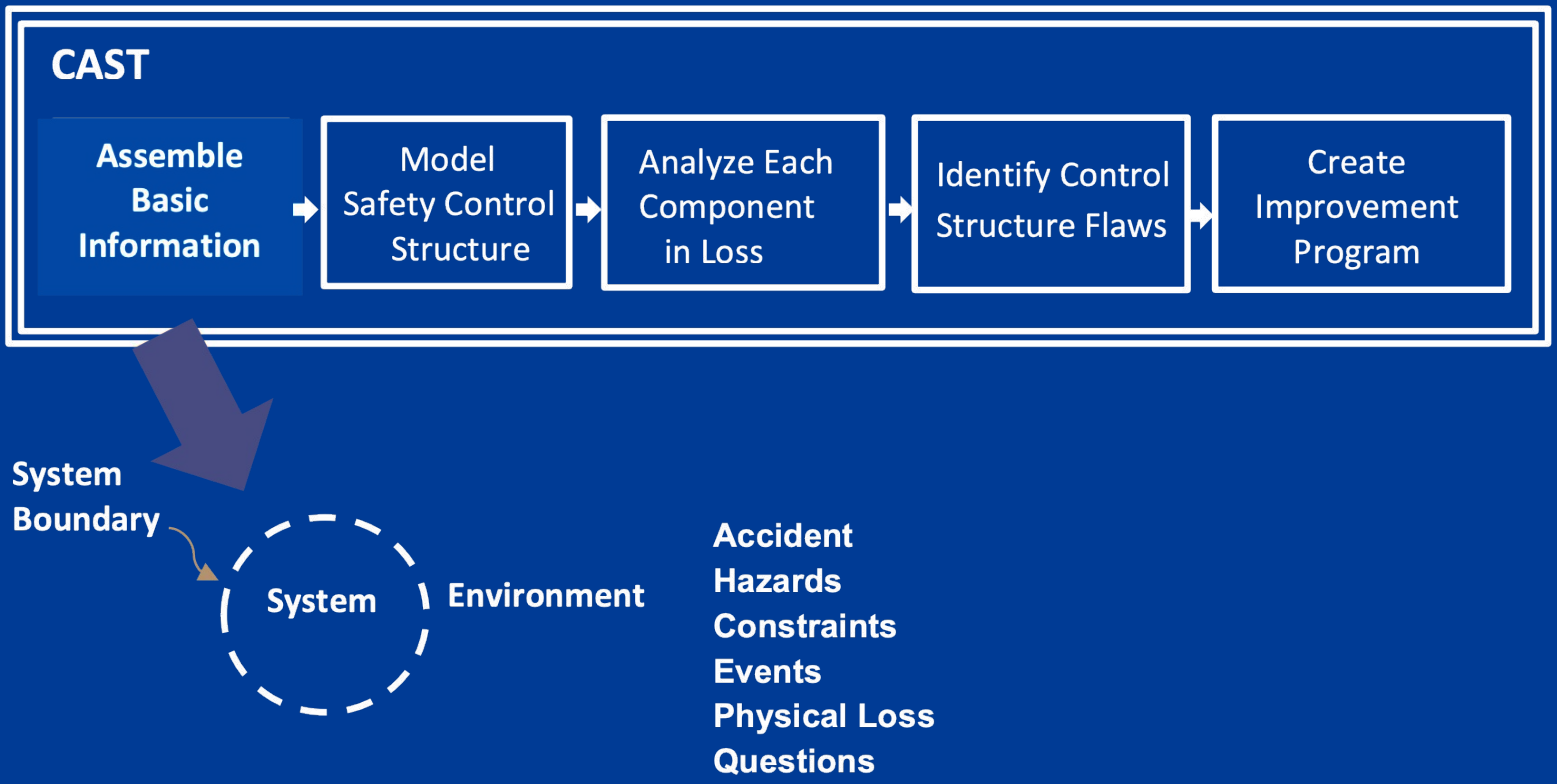

ID Need
* concept
* rqmts
Design
Create
* code
Install
* deploy
Operate
* maintain
* modify
* events
Uninstall
* Decomm
Dispose
STP
A
CAST
Accidents
Incidents
Near Miss
Accident
Analysis
Preliminary
Hazard Analysis
System and Subsystem
Hazard Analysis

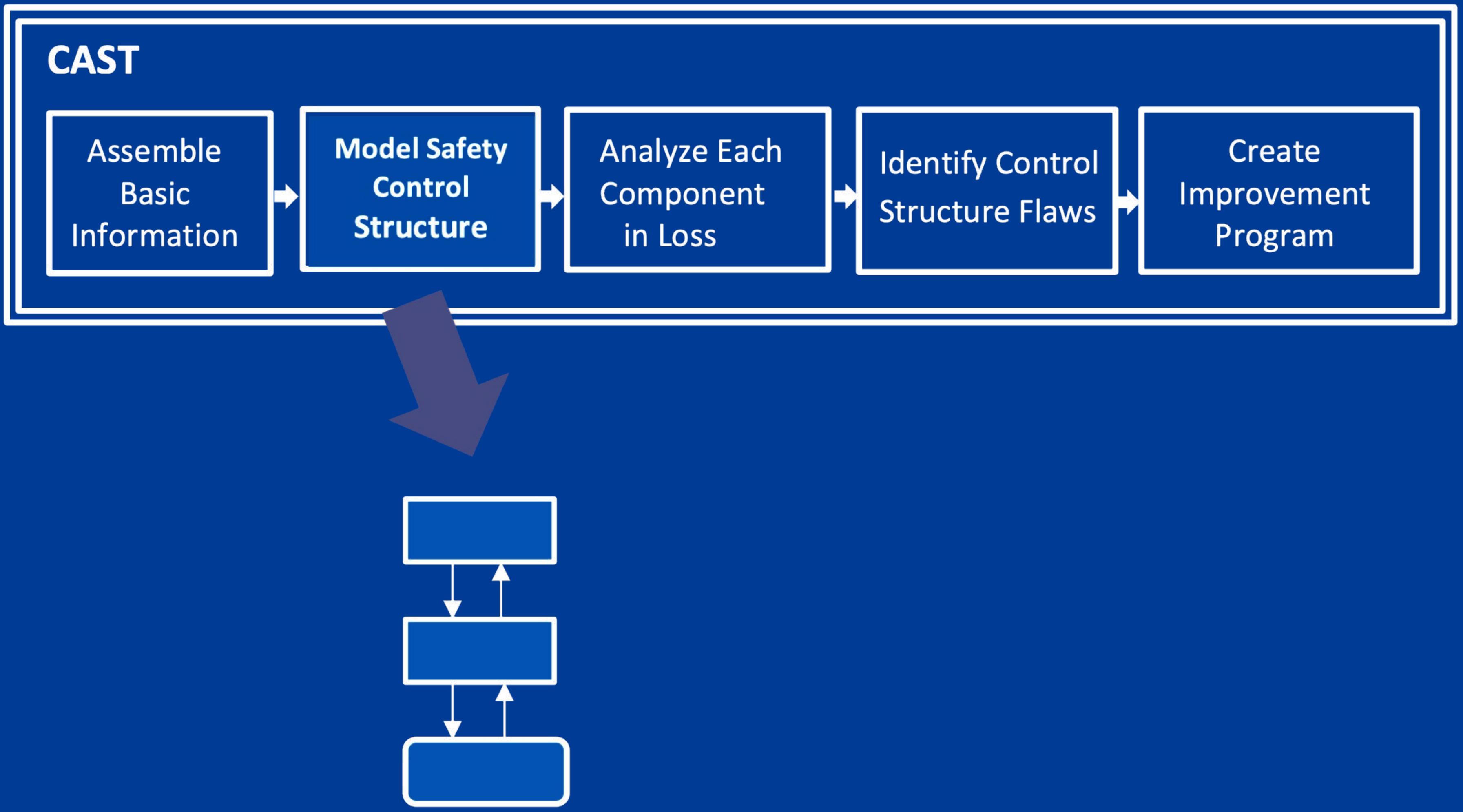
CAST as analysis method not an investigation technique

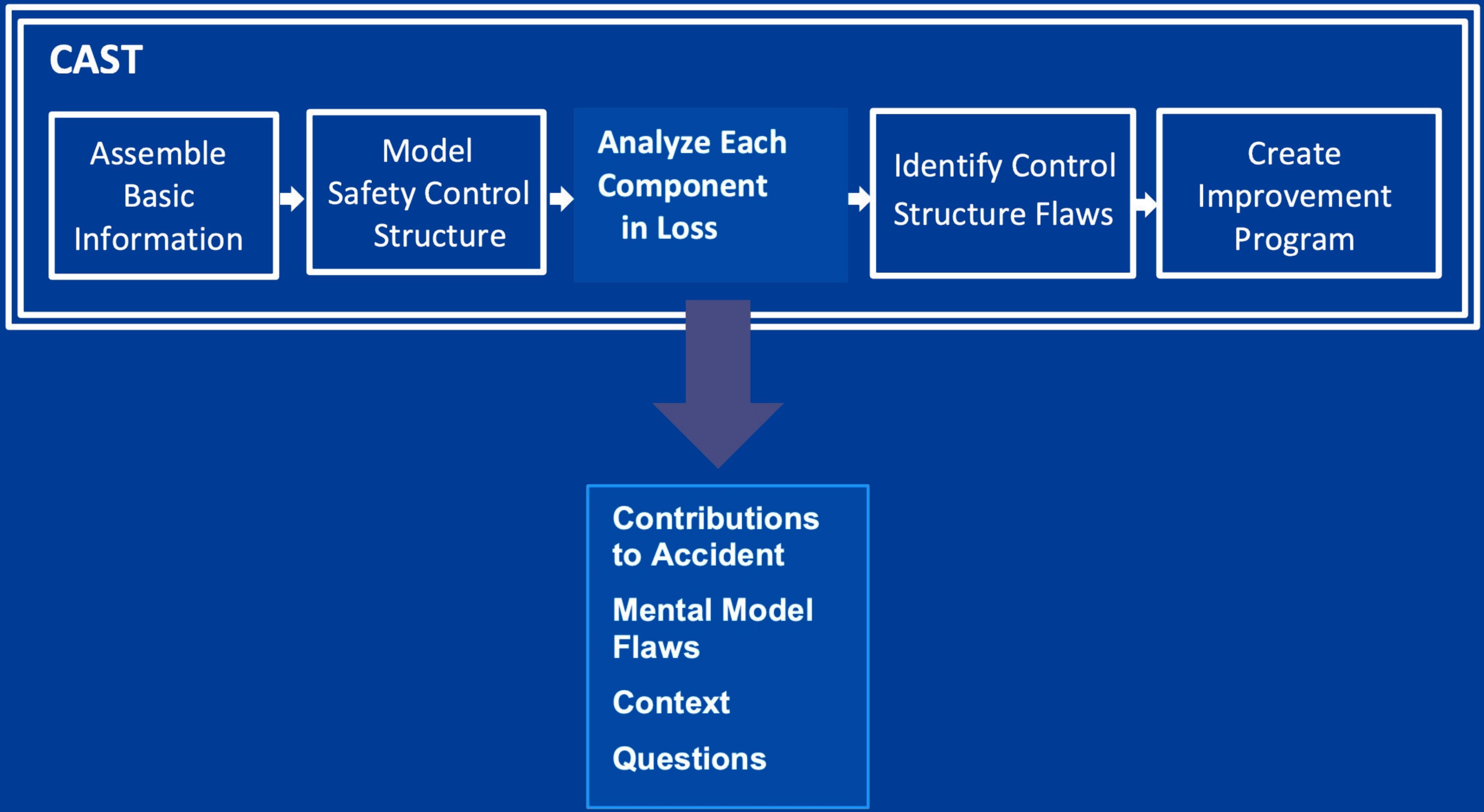
CAST goal: identify reasons why safety constraints were not enforced.

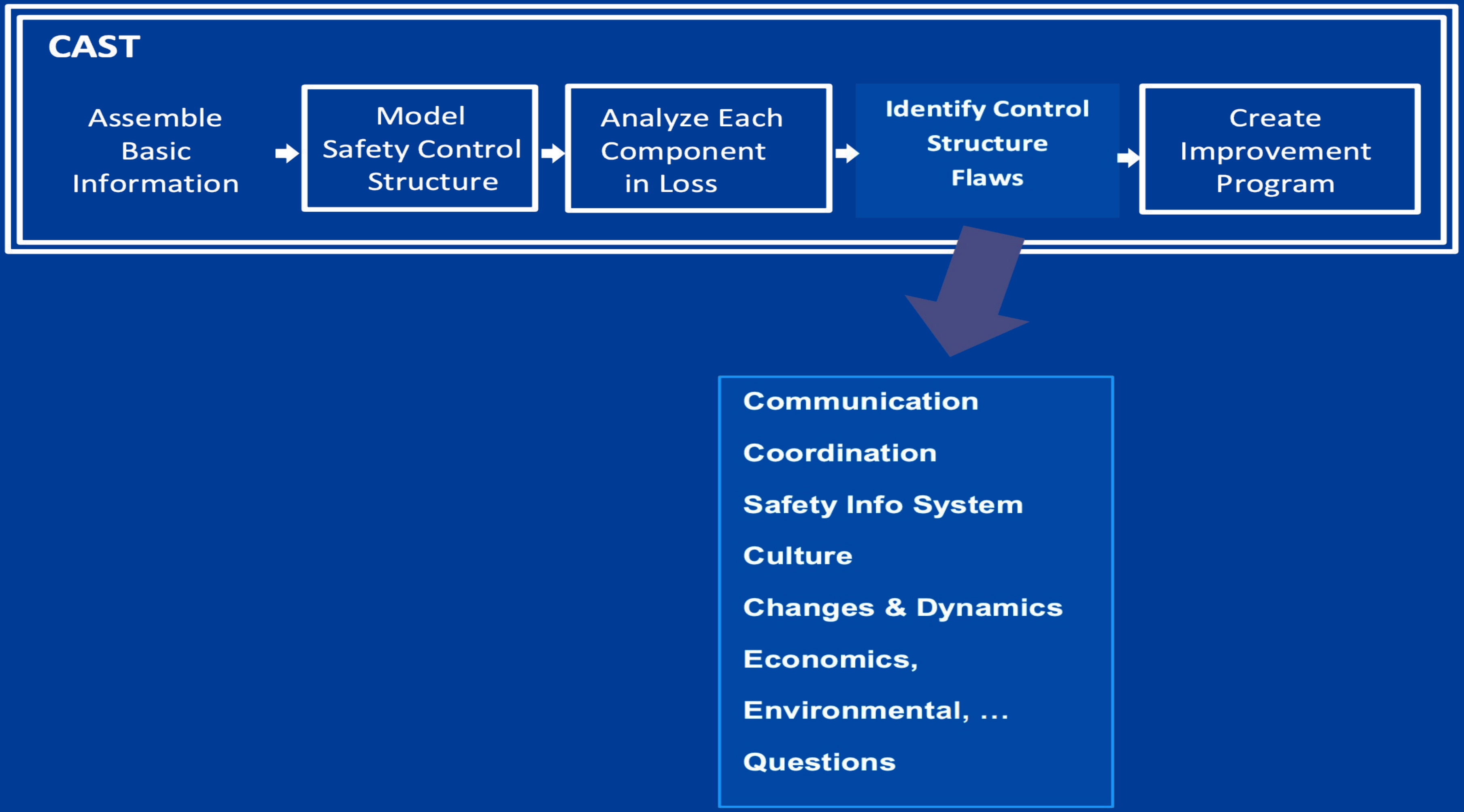
Analysis is performed in 5 parts:

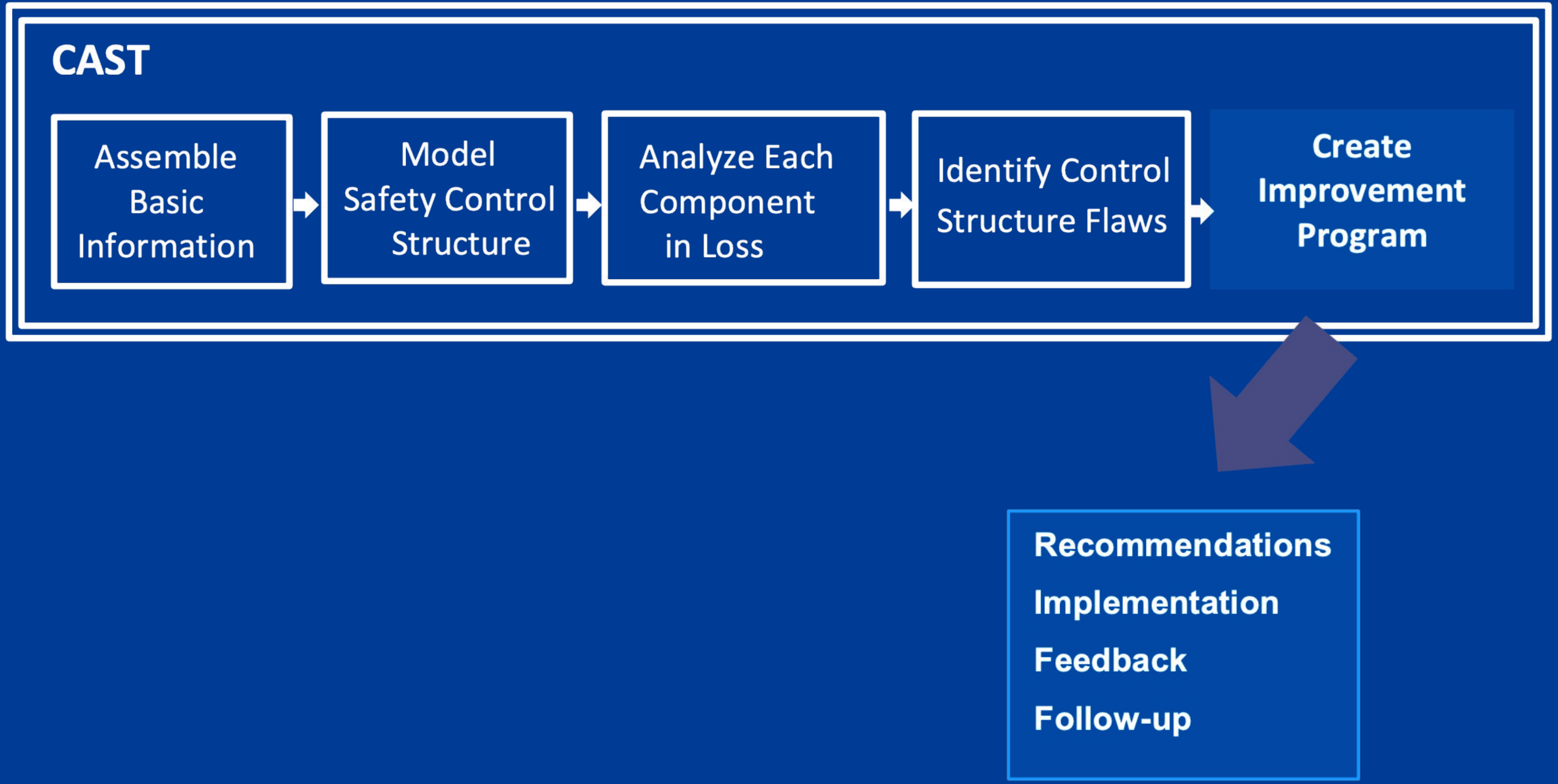
1. Assemble basic information
2. Model Safety control structure
3. Analyse Each component in Loss
4. Identify Control Structure Flaws
5. Create Improvement Program


CAST
Assemble Basic Information
Model Safety Control Structure
Analyze Each Component in Loss
Identify Control Structure Flaws
Create Improvement Program
System Boundary
System
Environment
Accident
Hazards
Constraints
Events
Physical Loss
Questions


CAST
Assemble Basic Information
Model Safety Control Structure
Analyze Each Component in Loss
Identify Control Structure Flaws
Create Improvement Program


CAST
Assemble Basic Information
Model Safety Control Structure
Analyze Each Component in Loss
Identify Control Structure Flaws
Create Improvement Program
Contributions to Accident
Mental Model Flaws
Context
Questions


CAST
Assemble Basic Information
Model Safety Control Structure
Analyze Each Component in Loss
Identify Control Structure Flaws
Create Improvement Program
Communication
Coordination
Safety Info System
Culture
Changes & Dynamics
Economics,
Environmental, …
Questions


CAST
Assemble Basic Information
Model Safety Control Structure
Analyze Each Component in Loss
Identify Control Structure Flaws
Create Improvement Program
Recommendations
Implementation
Feedback
Follow-up

Hang on! We are almost there

Why aren't we learning enough from accidents and incidents

CAST takeaways:

- Root cause seduction and oversimplification of causality
  - There could be Multiple root causes.
  - Why stop at one?

- Check why control structures were not effective
  - Was the design safe and did it match implementation?

- Unrealistic view of human error
  - The "Bad apple" theory is actually wrong!

Why aren't we learning enough from accidents and incidents

- Investigation should be blameless
  - blame is a legal/moral concept, not an engineering one.
  - Focus on "why" and "how", not "who"

- **HUMAN ERROR IS A SYMPTOM OF A "SYSTEM" THAT NEEDS A REDESIGN**
  - Take a system's view on human behaviour
  - Traditional thinking: Human error as a cause.

- Avoid Hindsight bias
  - Antipattern: "Should have, could have, would have" in the analysis

Overcoming Hindsight Bias

- None comes to work to do a bad job
  - Assume reasonable things were done, take in account complexity, dilemmas, tradeoffs and uncertainty surrounding them
  - Finding and highlighting mistakes does not explain why.
  - Saying what was done or not done does not explain why.

- Investigation reports should provide explanations
  - Why it made sense for an operator or a user to do what they did
  - What changes will reduce the chances of that happening again

References

- CAST handbook
  - http://sunnyday.mit.edu/CAST-Handbook.pdf
- CAST notes:
  - https://github.com/joelparkerhenderson/causal-analysis-based-on-system-theory
- Engineering a safe world:
  - http://sunnyday.mit.edu/safer-world.pdf
- STPA handbook
  - https://psas.scripts.mit.edu/home/books-and-handbooks/
- A systems approach to Safety and Cybersecurity
  - https://www.youtube.com/watch?v=LKI6oRZ49T0

“A learning experience is one of those things that says,
'You know that thing you just did? Don't do that.’ ”

Douglas Adams, The Salmon of Doubt, William Heinemann Ltd, 2001.